ARJUN ROSE J S

Email: arjunrose2005@gmail.com
Portfolio: https://arjunrose.github.io/Arjun_Rose_J_S/
Mobile: +91 8317312557
GitHub: https://github.com/arjunrose
LinkedIn: https://www.linkedin.com/in/arjun-rose-j-s

Summary

B.Tech Robotics and Automation engineer with hands-on experience in Festo MPS, Siemens PLC programming (TIA Portal), and digital manufacturing workflows. Proficient in integrating Industrial IoT, developing robotic prototypes, and validating mechanical systems using CAD-based digital twins. Experienced in working with industrial and collaborative robots including Yaskawa cobots and Motoman robots, as well as Igus Delta Robot and Dobot Magician.

Education

- **B.Tech Robotics and Automation (Specialization in Artificial Intelligence)** CGPA: 7.66
  Karunya Institute of Technology and Sciences, Coimbatore *2022–2026*

Professional Experience

- **Yaskawa India Private Limited - Internship** June 2025
  - Gained hands-on exposure to PLC-controlled industrial automation systems and Motoman MH24 (DX200) robot operations in a manufacturing environment. Studied Automatic Tool Changer (ATC) logic and workflow sequencing while understanding industrial robot commissioning practices.
- **Tamil Nadu Newsprint and Papers Limited - Internship** June 2024
  - Observed a large-scale plant automation architecture involving PLCs, ASRS, industrial sensors, and automated material handling systems. Analyzed system integration processes and gained a practical understanding of industrial control strategies in continuous manufacturing operations.

Technical Projects

- **Siemens PLC-Based Industrial Automation & Control**
  Programmed Siemens PLCs (Ladder Logic) using TIA Portal with HMI faceplates for automated systems. Configured I/O modules and executed software development lifecycle (SDLC) phases, including technical specifications and test processes, to simulate cost-efficient production methods.
- **Areca Palm Tree Climbing & Harvesting Robot**
  Designed and simulated a climbing robot in Autodesk Fusion 360, engineering motor enclosures, wheel geometry, and a spring-based gripper. Utilized generative design and motion links as a digital twin to validate physical mechanics.
- **Vision-Based Industrial Sorting Automation**
  Developed an automated sorting system with Python/OpenCV. Integrated sensor-triggered pneumatic actuation, performing diagnostics on signal timing and mechanical alignment to maintain high system accuracy.
- **Mimic Robotic Arm with Real-Time Control**
  Built a microcontroller-based robotic arm with real-time servo actuation via sensor input. Implemented control logic and signal processing for synchronized, stable industrial manipulation.

Technical Skills

Automation: Festo MPS, Pneumatics & Hydraulics, Siemens SIMATIC PLCs (TIA Portal), Ladder Logic

Industrial Systems: Industrial Robots (Yaskawa 6-Axis, Igus Delta, Cobots), ATC Logic

Programming & IIoT: Python, C, Industrial Communication (CAN), ESP32 Wroom 32 Wireless Networking, Edge Computing (Raspberry Pi, NVIDIA Jetson Nano)

Software & Tools: Siemens TIA Portal, Codesy, Autodesk Fusion, AutoCAD, SolidWorks, Linux, Git

Certifications & Online Courses

- **SIEMENS Certified Training Program on Basic Industrial PLC**
- Introduction to Industry 4.0 and Industrial IoT (NPTEL)
- Getting Started with AI on Jetson Nano (NVIDIA)
- Autodesk Fusion 360
- MATLAB Onramp (MathWorks)
- Network Addressing and Basic Troubleshooting (CISCO)
- Linux Tutorial (Great Learning Academy)